

ADVERSARIAL & SECURITY TESTING

Arabic Fake News Detection – Attack & Defense Analysis

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THE APPROACH

Treat the model as a security target, not just a classifier

\$ Why Adversarial Testing?

— The Real-World Threat

- ▶ Fake news creators are sophisticated – they WILL try to bypass your detector

— What we test for:

- ▶ Brittleness – can one changed character flip a 'Fake' label to 'Credible'?

- ▶ That is called an Inference Attack – and it is exploitable in production

- ▶ # # Models tested: ArBERT (formal news) vs MarBERTv2 (tweets)

- ▶ → 14 attack types · 600 samples · 2 models compared

KEY METRICS

What we measure to quantify security

\$ The 3 Security Metrics

METRIC

- ▶ AFR
- ▶ Adv-Acc
- ▶ Perturbation Sensitivity
- ▶ HCFR

WHAT IT ANSWERS

- ▶ % of correct predictions flipped by attack
 - ▶ Overall accuracy after the attack
- ▶ How much confidence shifts under attack
 - ▶ Confident AND wrong – the most dangerous failure

MODEL BASELINES

Where both models start – before any attack

\$ ArBERT vs MarBERTv2 – Baseline

ArBERT

- ▶ Trained on: formal Arabic news (AFND)
- ▶ Baseline accuracy: 95.5%
- ▶ Danger threshold: 5% AFR
- ▶ Critical threshold: 15% AFR
- ▶ Main weakness: character-level attacks

MarBERTv2

- ▶ Trained on: Arabic tweets (FakeArabTweets)
 - ▶ Baseline accuracy: 98.3%
- ▶ No attack crosses 5% AFR
- ▶ Zero high-confidence flips recorded
- ▶ Weakness: collapses on out-of-domain data

— PLOT 01 —

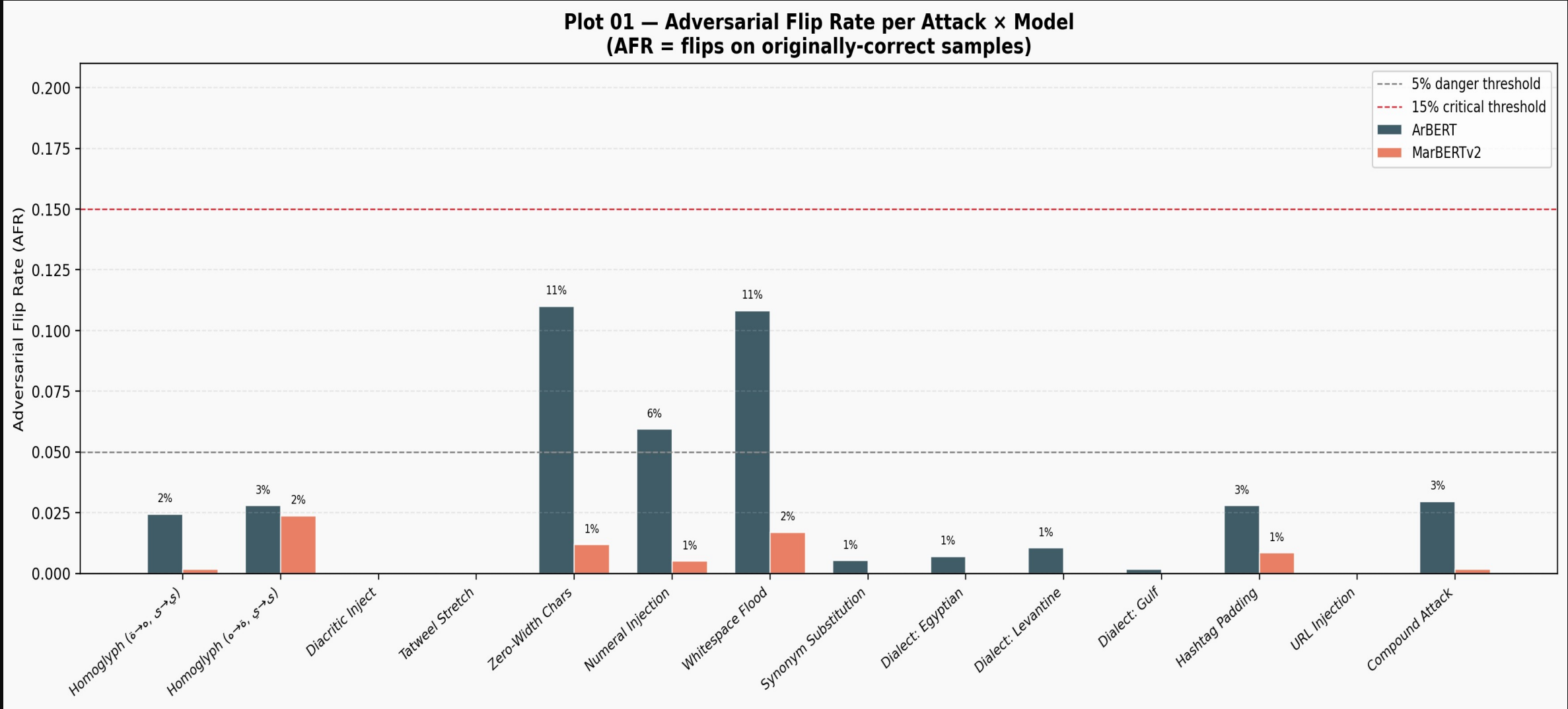
ADVERSARIAL FLIP RATE

Which attacks break each model – and by how much?

\$ Plot 01 – How to Read It

- ▶ AFR = out of all correct predictions, how many flipped to wrong after the attack?
 - ▶ Dark blue bars = ArBERT | Orange bars = MarBERTv2
- ▶ Grey dashed line = 5% danger threshold | Red dashed = 15% critical
- ▶ ArBERT: Zero-Width Chars and Whitespace Flood both hit 11% AFR
- ▶ Both cross the danger line – invisible characters and extra spaces break it
- ▶ MarBERTv2: max AFR only 2.4% – stays below danger line on all 14 attacks
- ▶ # # Tweet-trained models learn to ignore noisy formatting naturally

Plot 01 – Adversarial Flip Rate per Attack × Model



ArBERT: 11% AFR on Zero-Width Chars & Whitespace Flood

MarBERTv2: max 2.4%, never crosses danger line

— PLOT 02 —

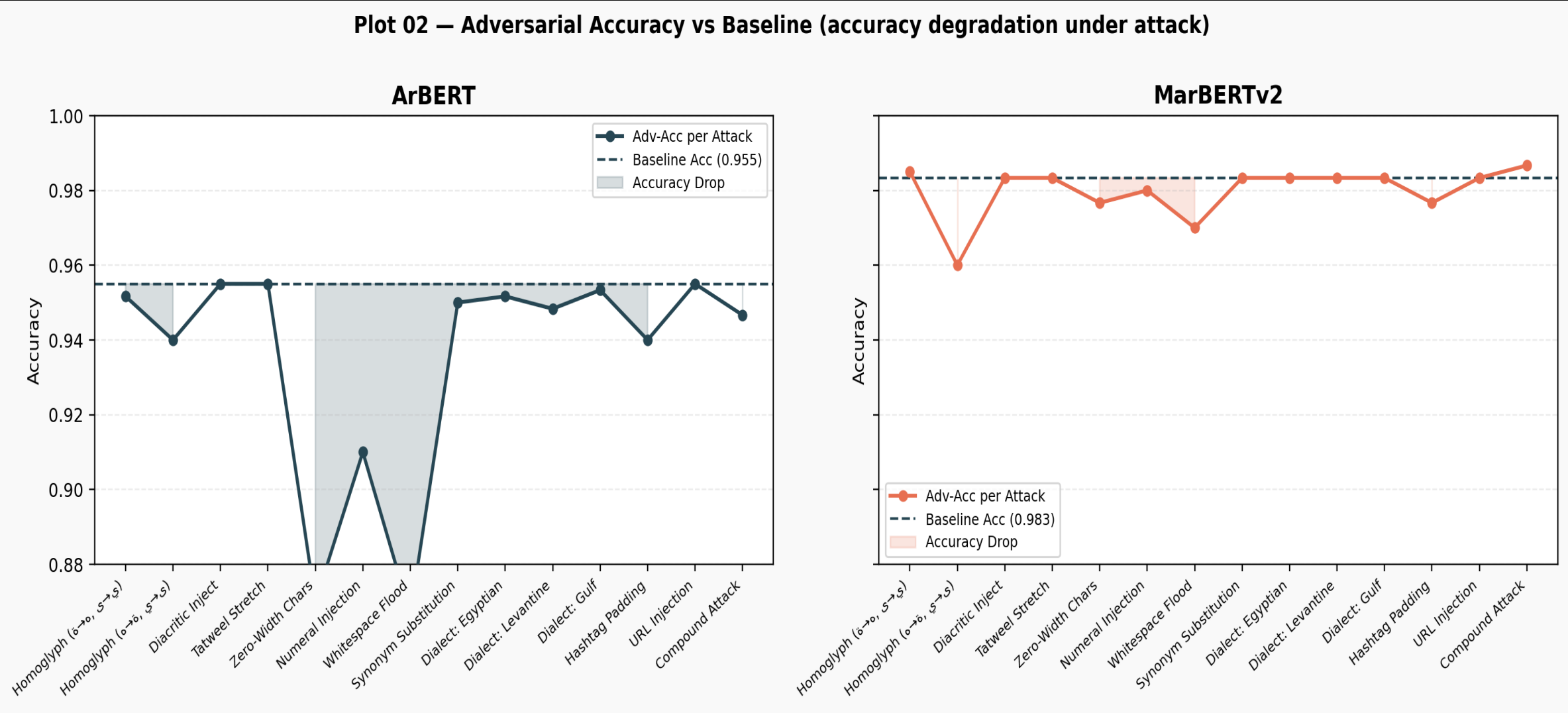
ADVERSARIAL ACCURACY VS BASELINE

Full accuracy picture under all 14 attacks

\$ Plot 02 – What the Accuracy Lines Show

- ▶ Dashed line = baseline accuracy (no attack) | Solid line = under attack
- ▶ ArBERT baseline: 0.955 → drops to 0.870 (Zero-Width) and 0.867 (Whitespace Flood)
- ▶ That is a drop of 8–9 percentage points – clearly visible shaded area
- ▶ MarBERTv2 baseline: 0.983 → biggest drop only 2.3 points (Homoglyph attack)
- ▶ Some attacks push MarBERTv2 slightly above baseline – within normal noise range
- ▶ # # ArBERT has 2 clear vulnerability points. MarBERTv2 has none.

Plot 02 – Adversarial Accuracy vs Baseline



ArBERT: two sharp dips at Zero-Width & Whitespace Flood | MarBERTv2: line barely leaves the baseline

— PLOT 03 —

PERTURBATION SENSITIVITY HEATMAP

Confidence changes – even when the prediction does not flip

\$ Plot 03 – Reading the Heatmap

▶ PS = how much the model's confidence changes under attack (even if label stays same)

▶ Dark red = high sensitivity (decision boundary is thin and fragile)

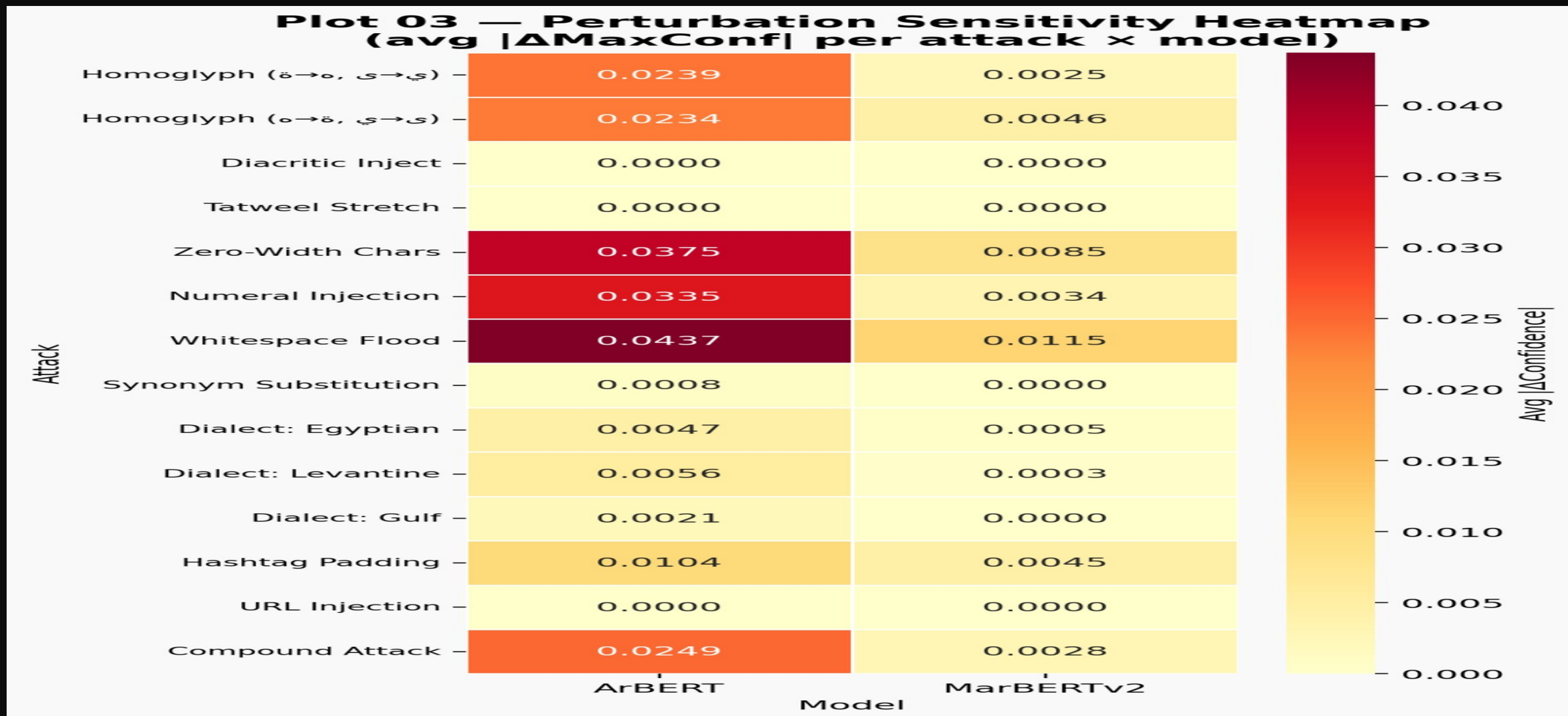
— ArBERT – top 3 danger cells:

▶ Whitespace Flood: PS=0.044 | Zero-Width Chars: PS=0.038 | Numeral Injection: PS=0.034

▶ MarBERTv2 – max PS=0.012 (Whitespace Flood) – nearly all yellow (stable)

▶ # # Diacritic Inject, Tatweel, URL = PS 0.000 on both – blocked by preprocessing

Plot 03 – Perturbation Sensitivity Heatmap



ArBERT: 3 dark red cells (Whitespace, Zero-Width, Numeral) | MarBERTv2: almost entirely light yellow

— PLOT 04 —

FLIP HEATMAP — SAMPLE LEVEL

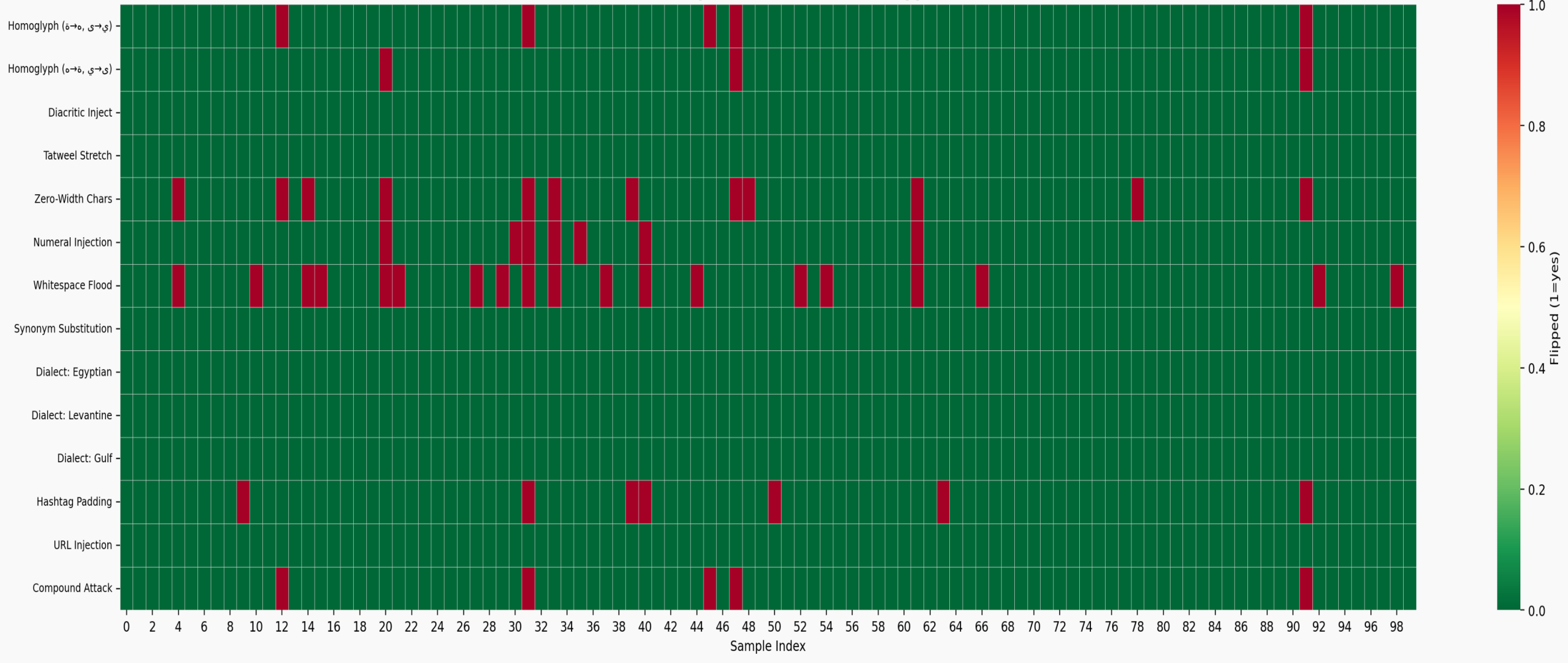
ArBERT vs MarBERTv2 — where exactly do flips happen?

\$ Plot 04 – How to Read the Heatmap

- ▶ X-axis = sample index (0-99) | Y-axis = attack type | Red cell = flipped
 - ▶ **ArBERT: Zero-Width and Whitespace** rows show many red cells spread across positions
 - ▶ Scattered reds = general vulnerability, not targeting one specific content type
 - ▶ Some samples appear red in BOTH rows – they are on the decision boundary for multiple attacks
 - ▶ **MarBERTv2: almost completely green** – only 3-4 isolated red cells total
 - ▶ # # Side-by-side, this is the strongest visual argument in the study

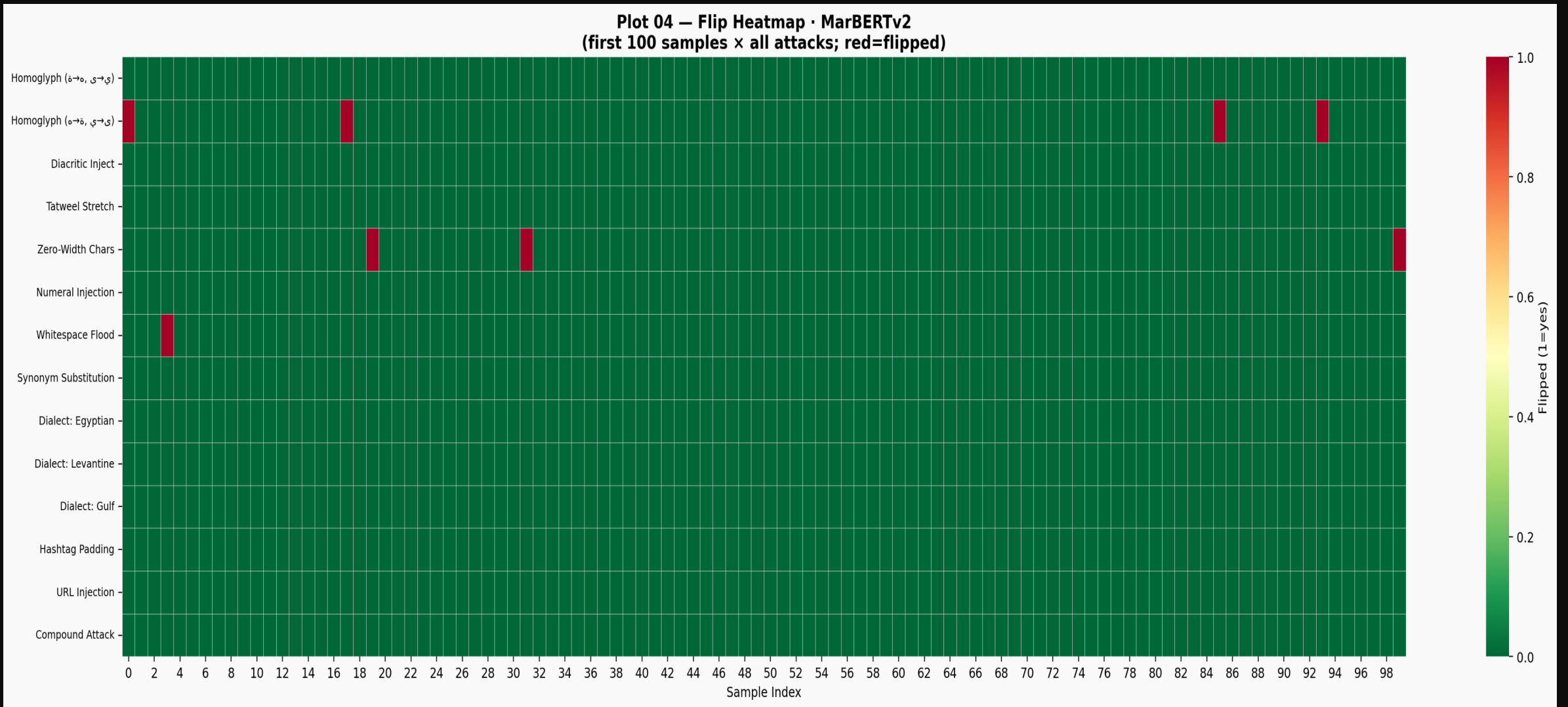
Plot 04 – Flip Heatmap · ArBERT (100 samples × 14 attacks)

Plot 04 – Flip Heatmap · ArBERT
(first 100 samples × all attacks; red=flipped)



Red patterns visible in Zero-Width and Whitespace rows – scattered across many sample positions

Plot 04 – Flip Heatmap · MarBERTv2 (100 samples × 14 attacks)



— PLOT 05 —

CLASS ASYMMETRY OF ATTACKS

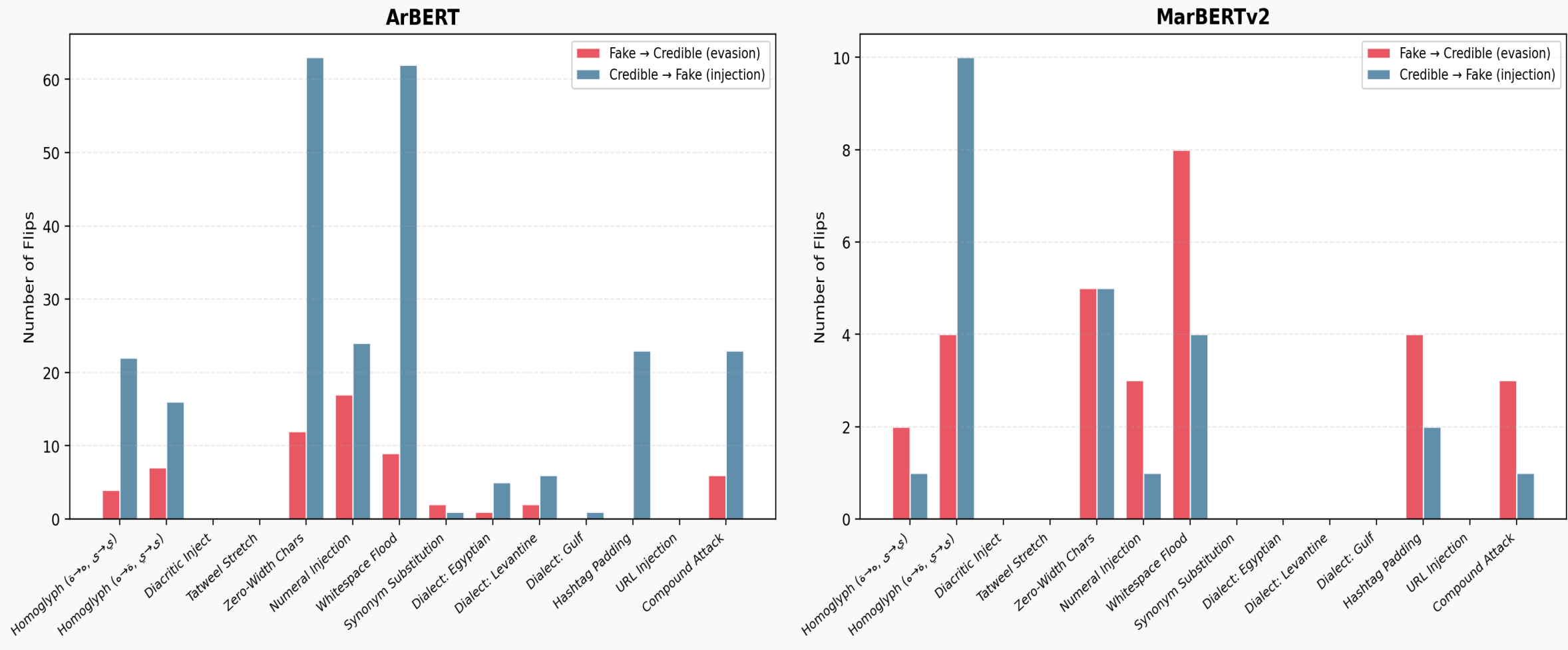
Evasion (Fake→Real) vs Injection (Real→Fake) – two different dangers

\$ Plot 05 – Two Directions of Error

- ▶ Red bar = Evasion: Fake slips through as Credible – misinformation spreads
- ▶ Blue bar = Injection: Real news flagged as Fake – legitimate content censored
- ArBERT – strong injection bias:
 - ▶ Zero-Width: 63 real→fake flips vs only 12 fake→real | Whitespace: 62 vs 9
 - ▶ Hashtag Padding: 0 fake→real flips, but 23 real→fake – ArBERT learned a wrong bias
 - ▶ MarBERTv2: both directions roughly balanced, much smaller bars overall
 - ▶ # # ArBERT can be exploited to wrongly flag legitimate news as fake

Plot 05 – Class Asymmetry of Attacks

Plot 05 – Class Asymmetry of Attacks
(evasion: fake→credible vs injection: credible→fake)



ArBERT: strong injection bias on Zero-Width and Whitespace attacks | MarBERTv2: balanced and small

— PLOT 06 —

HIGH-CONFIDENCE FLIP RATE

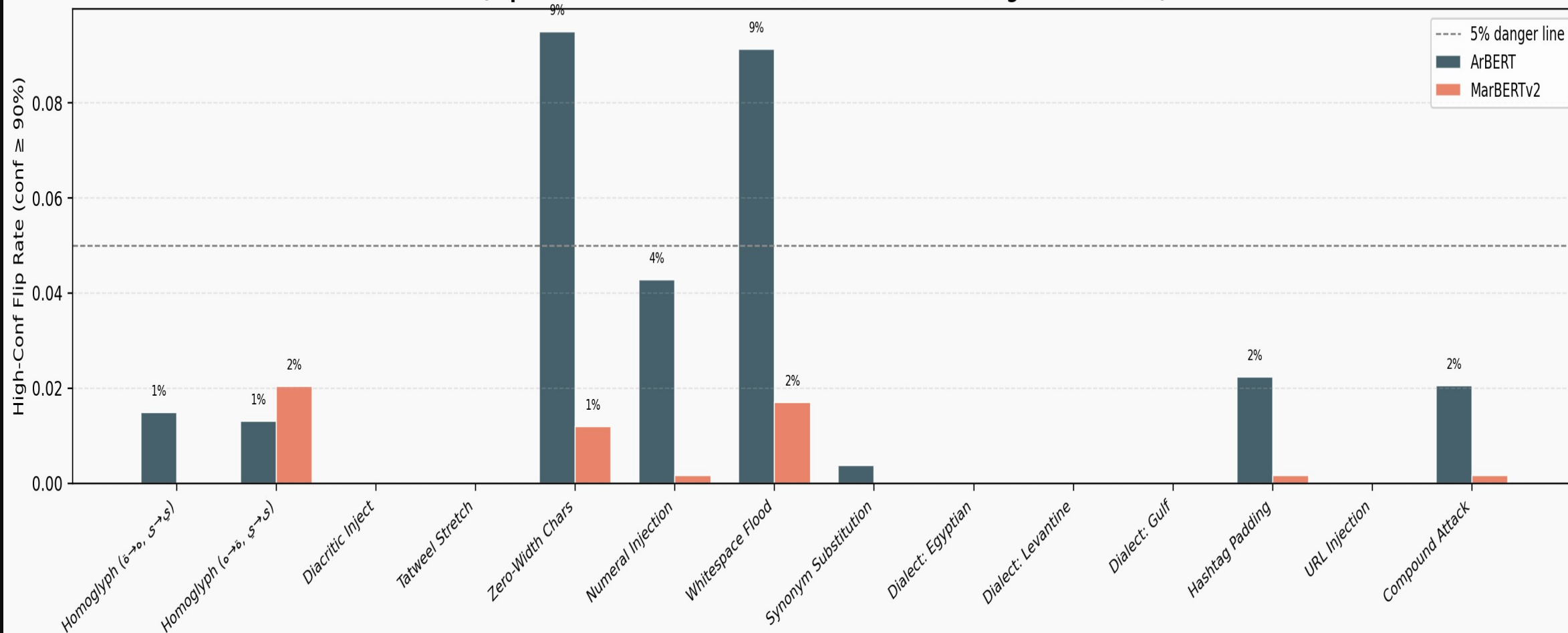
Wrong AND certain — the most dangerous failure mode

\$ Plot 06 – HCFR Explained

- ▶ HCFR = model was $\geq 90\%$ confident, but still got flipped to a wrong answer
- ▶ These errors will NEVER be flagged for human review – the system trusts itself
- ▶ ArBERT – above the 5% danger line:
- ▶ Zero-Width Chars: 9.5% HCFR | Whitespace Flood: 9.1% HCFR
- ▶ Numeral Injection also close at 4.3% – just below the line
- ▶ MarBERTv2: zero high-confidence flips across all 600 samples $\times$ 14 attacks
- ▶ # # HCFR of 0 is the strongest single robustness result in the study

Plot 06 – High-Confidence Flip Rate per Attack

Plot 06 – High-Confidence Flip Rate per Attack
(flips where model was $\geq 90\%$ confident $\rightarrow$ most dangerous failures)



ArBERT: 9.5% and 9.1% HCFR on Zero-Width and Whitespace – well above danger line | MarBERTv2: all near zero

— PLOT 07 —

DIALECTAL CAMOUFLAGE

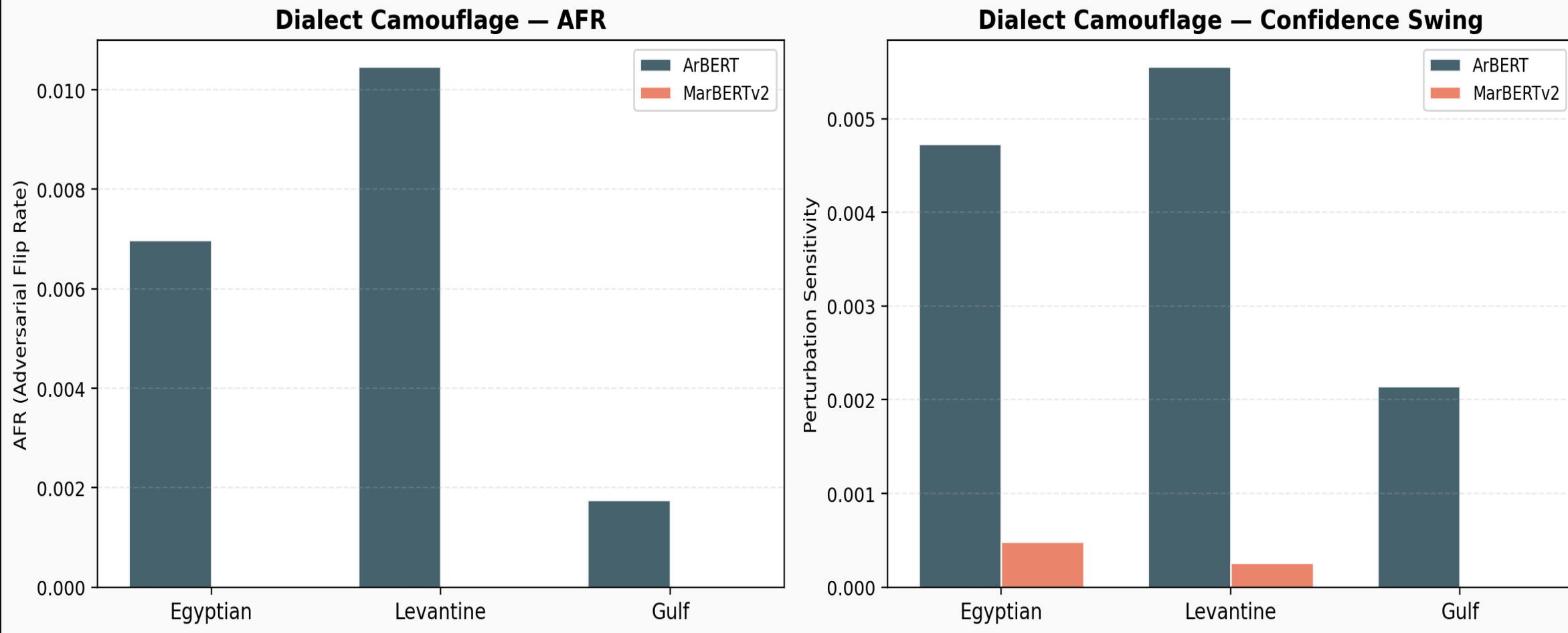
Same claim, different dialect – does the model care about style over substance?

\$ Plot 07 – The Style vs Substance Test

- ▶ Test: translate MSA text to Egyptian / Levantine / Gulf dialect – same meaning, different style
 - ▶ If model flips → it was judging style, not substance. That is a bias.
- ▶ ArBERT: Levantine shows highest AFR (~1.0%) and confidence sensitivity (PS=0.0056)
 - ▶ Still below the danger threshold – but more sensitive than MarBERTv2
- ▶ MarBERTv2: AFR and PS nearly zero for all 3 dialects – completely insensitive
 - ▶ # # Tweets naturally mix MSA and dialect – MarBERTv2 trained on that mix

Plot 07 – Dialectal Camouflage: AFR + Confidence Swing

Plot 07 – Dialectal Camouflage: MSA → Egyptian / Levantine / Gulf
(same claim, different register – style-over-substance bias test)



Both models pass the dialect test with low AFR | MarBERTv2 nearly unaffected – ArBERT slightly more sensitive

— PLOT 08 —

SANITY GATE DEFENSE

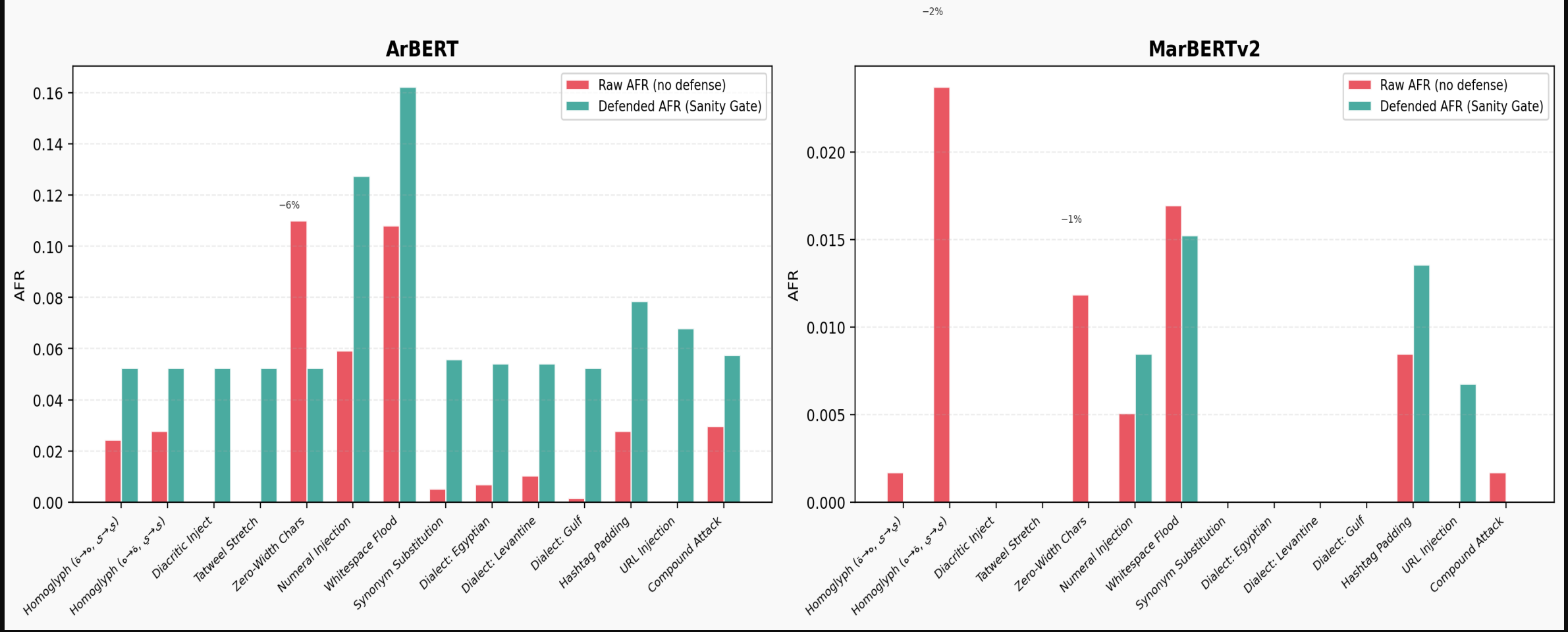
Does a normalization layer actually help?

\$ Plot 08 – When the Defense Backfires

- ▶ Sanity Gate = normalization layer: remove invisible chars, standardize Arabic letters, strip diacritics
 - ▶ ArBERT – the defense makes things WORSE:
- ▶ Diacritic Inject: 0% raw AFR → 5.2% defended | URL Inject: 0% raw → 6.8% defended
 - ▶ Reason: Defense-Induced Distribution Shift – ArBERT trained on ة and ى, gate removes them
 - ▶ Only Zero-Width Chars improved: 11% raw → ~5% defended (-6%)
- ▶ MarBERTv2: gate neutral or slightly helpful – Homoglyph AFR drops to near zero
 - ▶ # # Generic normalization cannot be applied blindly – must match training data

Plot 08 – Sanity Gate Defense: Raw AFR vs Defended AFR

Plot 08 – Sanity Gate Defense: Raw AFR vs Defended AFR
(normalization layer reduces attack success rate)



ArBERT: green bars taller than red on several attacks – defense backfires | MarBERTv2: neutral or slightly improved

— PLOT 09 —

COMPOUND ATTACK CASCADE

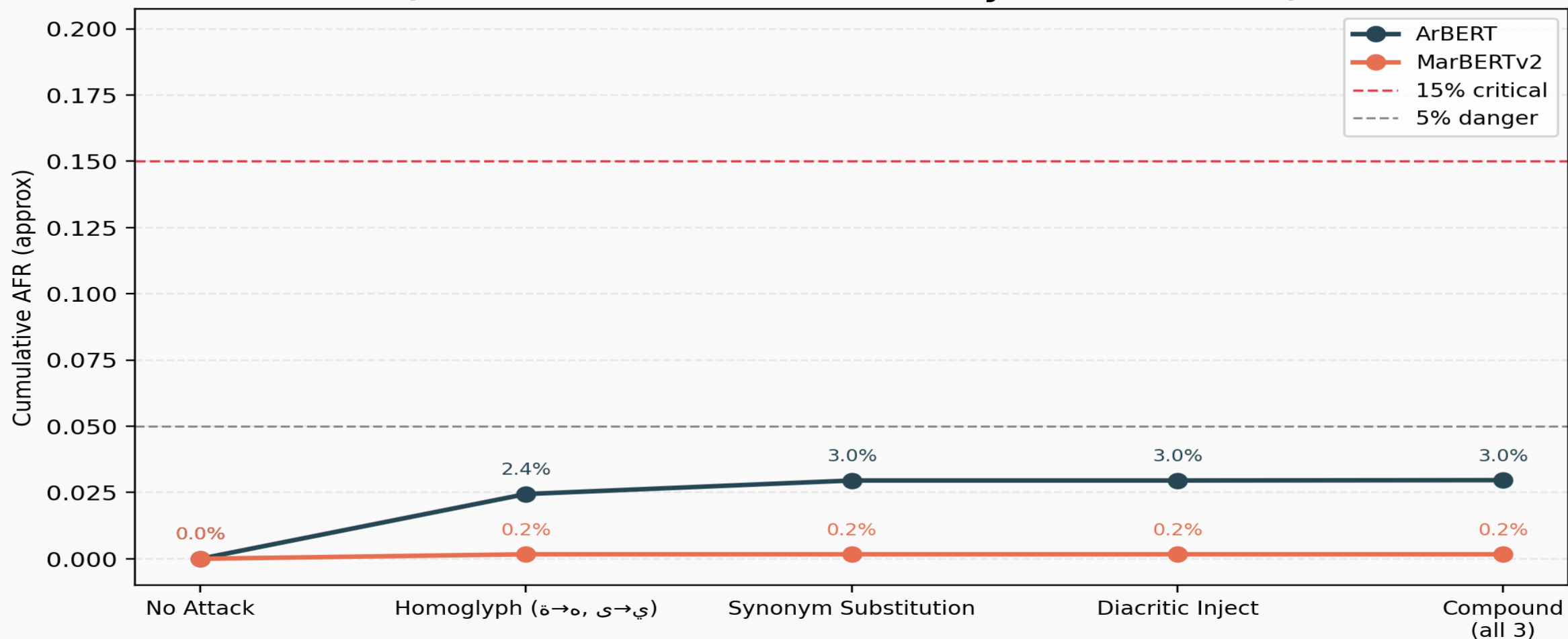
What happens when an attacker stacks multiple techniques?

\$ Plot 09 – Stacking Attacks in Layers

- ▶ Real attackers don't use just one technique – they combine several at once
 - ▶ Cascade: No Attack → Homoglyph → + Synonym Sub → + Diacritic Inject → all 3 combined
 - ▶ ArBERT: cumulative AFR goes 0% → 2.4% → 3.0% → 3.0% → stays at 3.0%
 - ▶ Low because the cascade avoids ArBERT's real weaknesses (Zero-Width, Whitespace)
 - ▶ MarBERTv2: stays at 0.2% even under the full compound attack
- ▶ # # Both models resist this specific cascade – but a smarter attacker could do worse

Plot 09 – Compound Attack Cascade

Plot 09 – Compound Attack Cascade
(how AFR escalates as attack layers are stacked)



ArBERT reaches 3.0% after all 3 layers | MarBERTv2 stays at 0.2% | Both below the 5% danger line

— PLOT 10 —

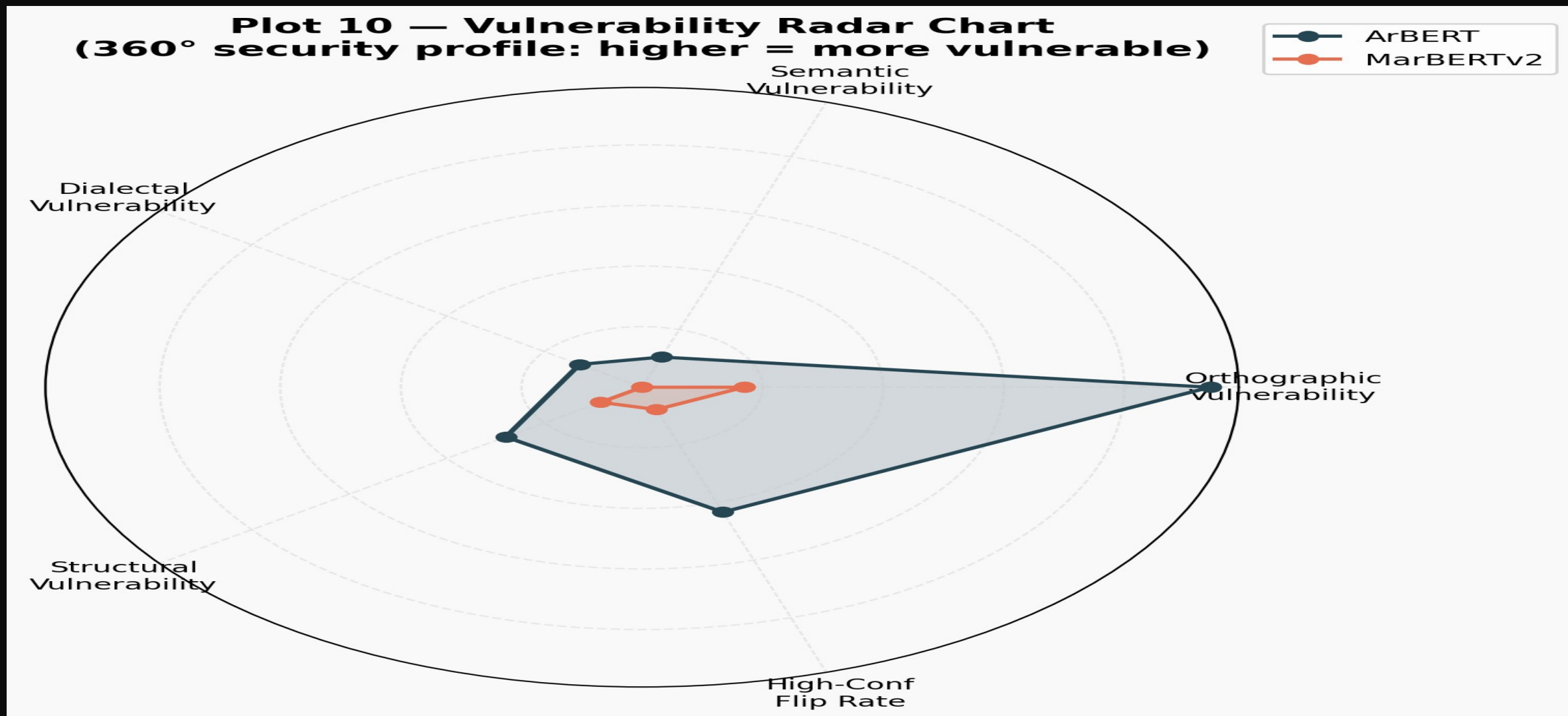
VULNERABILITY RADAR CHART

Complete 360° security profile – one view, five dimensions

\$ Plot 10 – Reading the Radar

- ▶ 5 security dimensions: Orthographic · Semantic · Dialectal · Structural · HCFR
- ▶ Further from center = MORE vulnerable on that dimension
- ▶ ArBERT: large dark blue polygon – especially stretched on Orthographic and HCFR
- ▶ Confirms all previous plots – character-level and confident-wrong are the weak points
- ▶ MarBERTv2: small compact orange shape near center – low on all 5 dimensions
- ▶ # # Best single slide to show a non-technical audience which model is safer

Plot 10 – Vulnerability Radar Chart (360° Security Profile)



ArBERT: large polygon, especially on Orthographic and HCFR | MarBERTv2: compact shape near center

— PLOT 11 —

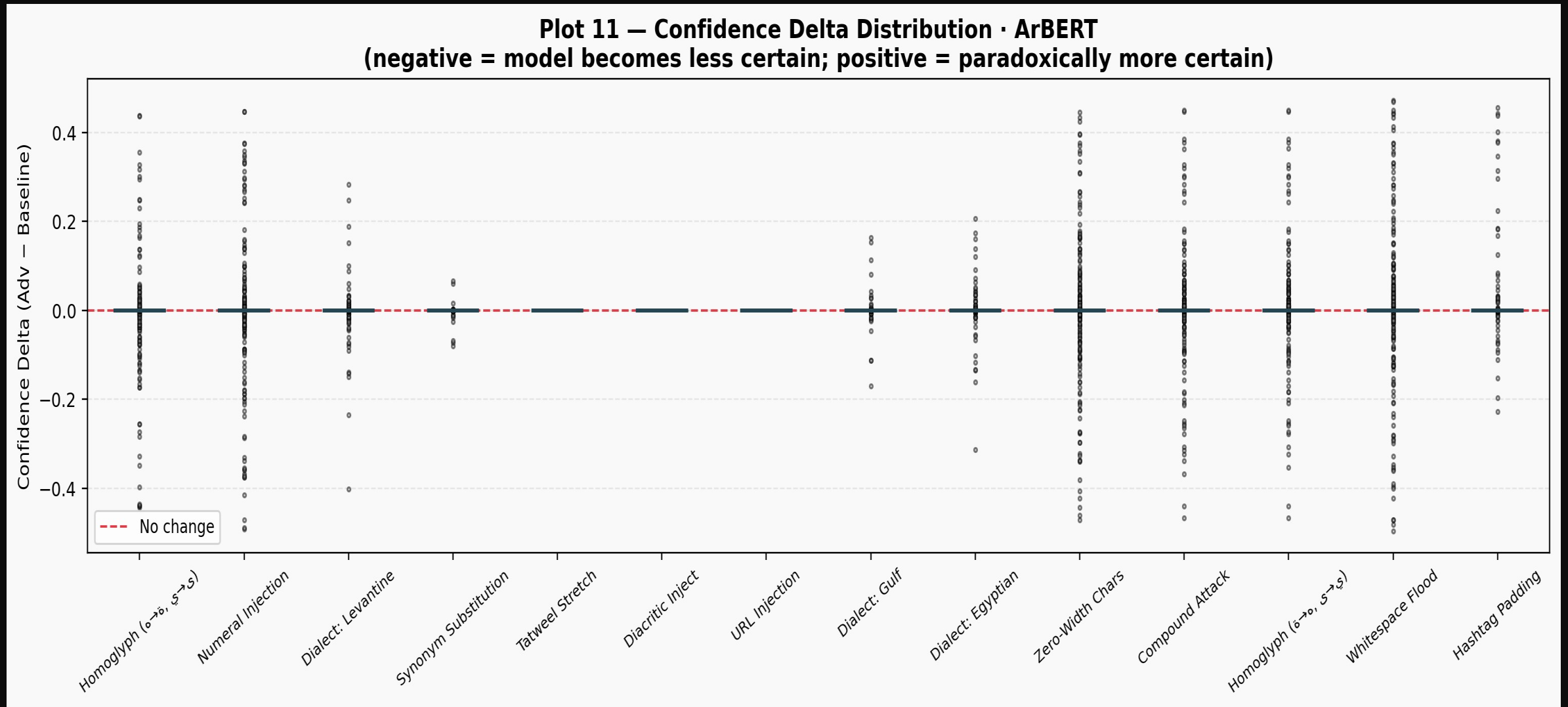
CONFIDENCE DELTA DISTRIBUTION

ArBERT vs MarBERTv2 – internal certainty under attack

\$ Plot 11 – What Confidence Delta Means

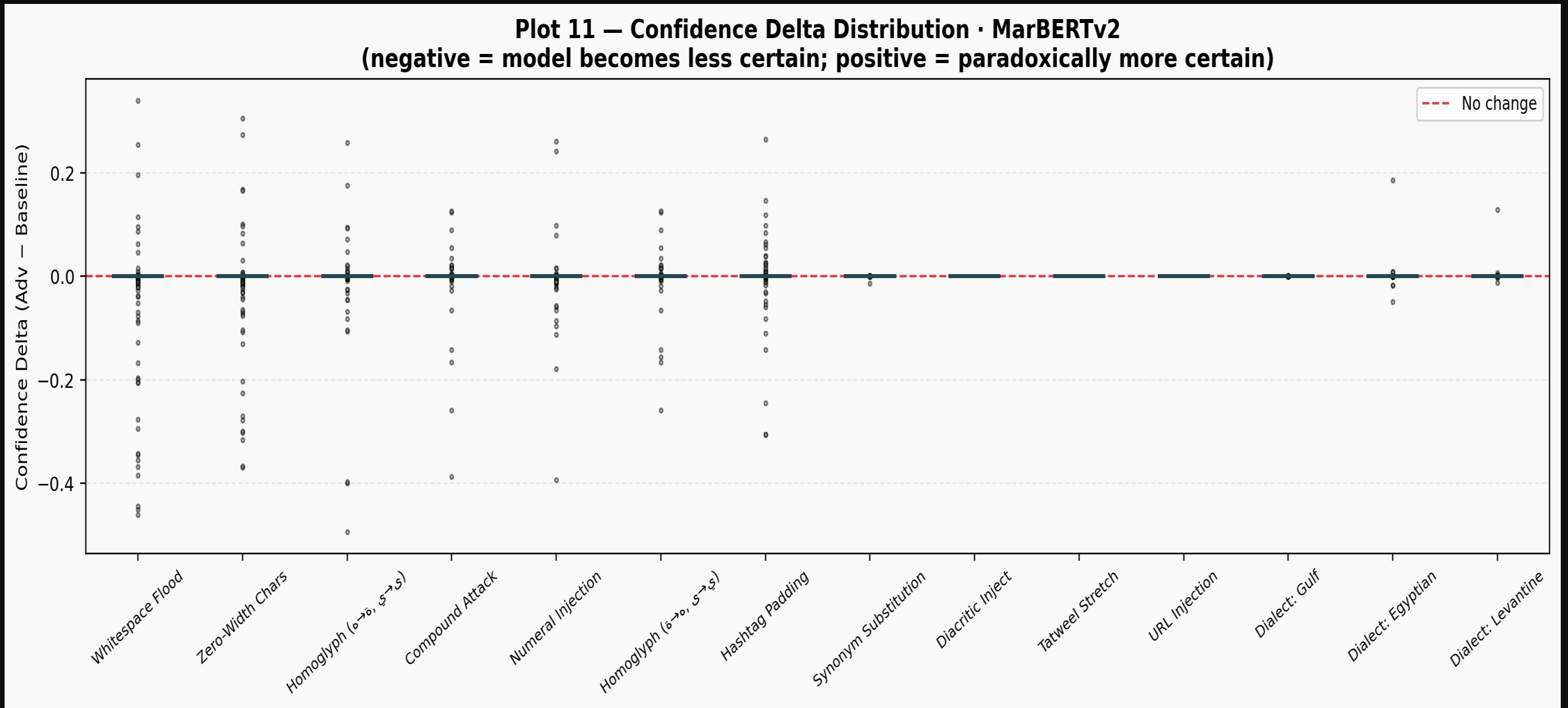
- ▶ Delta = confidence after attack minus confidence before attack
- ▶ Zero (red dashed line) = no effect | Negative = less certain | Positive = paradoxically MORE certain
- ▶ ArBERT: wide scatter on Zero-Width, Whitespace, Numeral – outliers reach ± 0.45
- ▶ Positive deltas are dangerous: model flips AND becomes more sure about the wrong answer
- ▶ MarBERTv2: almost all points clustered tightly at zero across all attacks
- ▶ # # Comparing both plots side by side is the clearest visual in the entire study

Plot 11 – Confidence Delta Distribution · ArBERT



Wide scatter on key attacks – some positive deltas: model flips and becomes MORE confident in wrong answer

Plot 11 – Confidence Delta Distribution · MarBERTv2



Nearly all points at zero – tight band across all 14 attacks | MarBERTv2 confidence is internally stable

— CONCLUSION —

FINAL VERDICT

Security vs Generalizability – the core tradeoff

\$ Key Takeaways

- ArBERT — strong baseline, real exploitable vulnerabilities
 - ▶ Zero-Width Chars and Whitespace Flood: 11% AFR, 9.5% HCFR — both above danger line
 - ▶ Strong injection bias: can be exploited to censor legitimate news
 - ▶ Sanity Gate defense backfires due to distribution shift
- MarBERTv2 — near-perfect robustness, but domain-locked
 - ▶ Zero AFR crossings, zero high-confidence flips across 600 samples × 14 attacks
 - ▶ BUT: collapses to 51.5% accuracy on out-of-domain formal news data
 - ▶ # # Future work: domain-invariant, adversarially robust Arabic NLP models

[EOF]

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